Panyi Dong

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EDUCATION

University of Illinois at Urbana-Champaign (UIUC) **Champaign, Illinois, US**

PhD, Mathematics focusing on Actuarial Science *Aug 2022-Present*

Research Focus: Data Science Applications in Actuarial Science, Machine Learning, Deep Learning

University of Illinois at Urbana-Champaign (UIUC) **Champaign, Illinois, US**

Master of Science, Mathematics focusing on Actuarial Science *Aug 2020-May 2022*

Relevant Courses: Actuarial Statistics, Extreme Value Theory and Reinsurance, Statistics Learning, Deep Learning, Loss Data Analytics & Credibility, Actuarial Research

GPA: 4.0/4.0

Zhejiang University (ZJU) **Hangzhou, Zhejiang, China**

Bachelor, Physics *Sep 2016-Jun 2020*

Relevant Courses: Mathematical Analysis I & II, Methods of Mathematical Physics I & II, Computational Methods, Ordinary Differential Equations, Theoretical Mechanics, Electrodynamics, Thermodynamics and Statistical Physics, Quantum Mechanics I & II, Experimental Economics, Life Contingencies

- **Honors and Awards:** Second-Class Scholarship for Elite Students in Basic Science, Certificate of Chu Kochen Honor Program

PROFESSIONAL EXPERIENCE

AIA Life Insurance Company Limited, Actuarial Department **Shanghai, China**

Intern, Reporting Team, Actuarial Department *Jan 2021-Jun 2021*

- Help team with transferring database management from FoxPro to Oracle SQL Database
- Help automating IBNR production procedure under IFRS/CGAAP/STAT accounting standards using Oracle and Python (This work has been presented at an internal sharing course)
- Help with additional requirements for IFRS17 reporting procedure and other daily work

Deloitte Consulting, Consulting Department **Hangzhou, China**

Intern, Online *Aug 2019-Sep 2019*

- Collect operating profits, stock price and sells volume data of automobile-dealer companies to analyze the transformation of automobile market and development of individual companies
- Analyzed the automobile manufacturers' marketing strategy and brand management
- Using the correlation of manufacturers' strategies and variation in sales volume to decide a suitable and profitable marketing tactic

RERSEARCH EXPERIENCE (https://github.com/PanyiDong)

Auto Machine Learning/AutoML (IRisk Lab, UIUC) *Sep 2021-Now*

- Read references on AutoML procedures and learn specific methods in each step of AutoML

- Write our own AutoML procedures with special focus on insurance data
- Looking for more ML models, test performances of different methods in data preprocessing and training
- Build an auto pipeline for data preprocessing and model selection/optimization procedures.

Extreme Precipitation Analysis and Quantifying Flood Damage (UIUC) *Sep 2020-Dec 2020*

- Fitting extreme precipitation data of Spain to Generalized-Pareto distribution
- Applied multiple regression model for spatial probability distribution of extreme values with geographic and topographic dataset
- Apply St. Venant model for overland flows for further quantifying flood damage work
- Trying to incorporate market-based insurance feedback and adaptation ideas to reduce the flood impact

Polymer Chains Driven Translocation through Pores (ZJU) *Aug 2019-Jun 2020*

- Constructed report of experimental, theoretical and simulation researches conducted in the polymers translocation topic and completed the code designing
- Completed programming and Monte Carlo dynamic sampling statistics algorithm
- Completed polymer escape simulation and compared the results with translocation statistics, numerically analyzed the physics properties

Viscoelasticity of Polymer (ZJU) *Mar 2019-Jun 2019*

- Learnt the significant polymer properties, creep and stress relaxation, and determined the advantages and setbacks using classical models (like Maxwell and Voigt model) to explain these two phenomena
- Based on Hookean bead-spring chains, employed Ito-type Fokker-Planck motion equation and introduced stochastic Brownian force to iterate and simulate the evolution process of polymers by calculating conformations and velocity fields
- Investigated the evolution properties of multiple polymers system

Glass Transition of Colloid System (ZJU) *Sep 2018-Dec 2018*

- Read related theses to learn theoretical concepts and simulation methods of soft matter/polymer systems
- Designed algorithm to simulate motion of system contained hard-sphere molecules using Monte Carlo method and investigated the correlation of molecular volume and cage effect during glass transition to determined glass transition critical conditions
- Used periodic boundary to simulate actual colloid system and employed cell-list method to reduce calculation resources

PERSONAL STATEMENT

- I have been fascinated with the idea of bringing automation using machines and innovations in statistics, Machine Learning to previous manual work to improve both efficiency and performance. With the current both academic/industrial trend in Neural Network and acceleration brought by the impact of the global pandemic, exploding demands for Big Data, Artificial Intelligence, our knowledge in risk management will be profoundly influenced. I wish to get deeply involved in the development of Data

Science and bring innovation in statistics methods to insurance and risk management world and explore those areas previously limited by knowledge.

SKILLS

- **Certificates:** SOA (Society of Actuaries): Exam P (Probability), FM (Financial Mathematics), IFM (Investment and Financial Mathematics), LTAM (Long-Term Actuarial Mathematics), STAM (Short-Term Actuarial Mathematics), VEE Mathematics Statistics
- **Programming and Software:** C, Matlab, VBA, Microsoft Office, Python, Fortran, LaTeX, Markdown
- **Language:** TOFEL 112(30+30+24+28), GRE 148+170+4